Type 1: Research Project

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Bi Temporal Satellite Image Change Detection Using Deep Learning Frameworks

23CSE498 – Project Phase 2 (Panel Review 1) | **Team Number: 63**

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Introduction

- **Change Detection:**
  - Identifies differences in land cover using satellite images from different time periods.
  - Examples: urban expansion, deforestation, and floods.
- **Types of Change Detection:**
  - **Short-term:** Captures rapid changes such as floods, fires, and construction.
  - **Long-term:** Captures gradual changes such as urban expansion and deforestation.
- **Change Analysis:**
  - Interprets detected changes to understand what changed, where it occurred, and its impact.
- **Bi-Temporal Change Detection:**
  - Compares two images acquired at different time points, T_1 and T_2.
  - Detects changes using spatial and spectral differences.
- **Multi-Temporal Change Detection:**
  - Uses images from multiple time points to analyze changes and trends over time.

Literature Review Comparison Table (1/2)

Title	Datasets	Year	Methodology	Accuracy	Gaps
ChangeFormer: A Transformer-Based Siamese Network for Change Detection	LEVIR-CD, DSIFN-CD	2022	Siamese transformer extracts multi-scale features from bi-temporal images and decodes them to produce pixel-level change maps.	LEVIR-CD Accuracy: 96% , DSIFN-CD Accuracy: 92–93%	Requires large training data and weak temporal modeling.
BIT: Remote Sensing Image Change Detection with Transformers	WHU-CD, LEVIR-CD	2021	CNN features are converted to tokens and processed with transformer attention to model temporal differences.	LEVIR-CD Accuracy: 96.1% , WHU-CD Accuracy: 95.8–96.0%	High computational complexity for large images.
FCCDN: Feature Constraint Network for VHR Image Change Detection	LEVIR-CD, WHU-CD	2021	Siamese CNN architecture with feature constraint learning and multi-scale feature pyramid networks for improved feature extraction and change detection.	LEVIR-CD F1 Score: 92.29% , WHU-CD F1 Score: 93.73%	Requires large pixel-level labeled datasets.
Self-supervised Remote Sensing Images Change Detection at Pixel-level	OSCD, MUDS	2021	Self-supervised Siamese deep learning model using contrastive learning to generate pixel-level change maps from bi-temporal images.	OSCD Accuracy: 89-90%	Requires large datasets and is sensitive to seasonal variations, limiting the detection of fine-grained changes.

Literature Review Comparison Table (2/2)

Title	Datasets	Year	Methodology	Accuracy	Gaps
AFCF3D-Net: Adjacent-Level Feature Cross-Fusion Network	WHU-CD, LEVIR-CD	2023	Takes two time images (T1, T2). Combines them using 3D conv to learns change across time	WHU-CD, LEVIR-CD Accuracy: 97%	High GPU memory usage and limited to buildings change detection.
A2Net: Lightweight Remote Sensing Change Detection Network	WHU-CD, LEVIR-CD	2025	Look at image in different zoom levels. Combine them to detect changes	WHU-CD, LEVIR-CD Accuracy: 96%	Difficulty detecting small buildings and subtle changes.
DMINet: Dual-Branch Multilevel Intertemporal Network	WHU-CD, LEVIR-CD	2021	Dual-branch network with intertemporal attention captures temporal feature differences.	WHU-CD, LEVIR-CD Accuracy: 96%	Sensitive to illumination and seasonal variations.
SNUNet-CD, STANet: Remote Sensing Image Change Detection with Transformers	LEVIR-CD, WHU-CD	2020	Dense skip connections enhance multi-scale feature propagation in Siamese UNet.	LEVIR-CD, WHU-CD Accuracy: 95–96%	High computational cost and boundary inaccuracies.
IFNet: Interactive Feature Fusion Network	LEVIR-CD, OSCD	2025	Interactive feature fusion modules exchange temporal information between bi-temporal branches for change detection.	LEVIR-CD, OSCD Accuracy: 95%	Limited ability to distinguish seasonal variations from real structural changes.
TinyCD: A (Not So) Deep Learning Model For Change Detection	LEVIR-CD, WHU-CD	2022	Lightweight Siamese U-Net exploiting low-level features, combined with a Mix and Attention Mask Block (MAMB) for space-time feature fusion at very low computational cost.	LEVIR-CD F1 Score: 91.05% , WHU-CD F1 Score: 91.74%	Lower semantic depth than transformer-based models; accuracy trails heavier SOTA methods on complex scenes.

Problem Statement

Change detection using satellite imagery identifies land cover variations over time by comparing bi-temporal images. However, accuracy is challenged by noise, spectral similarity, illumination differences, and class imbalance. These challenges necessitate robust deep learning methods for effective feature learning and improved detection.

Technical Knowledge & Design Decisions

Motivation from SOTA Analysis and Design Choices

Inspired by Tiny-CD

- **Tiny-CD** – best-performing lightweight model in our SOTA comparison.
- Strong accuracy with very **few parameters** (0.285M) and **low GFLOPs** (3.088).
- Proved **efficiency and accuracy can coexist** – inspired CAST-CD's design.

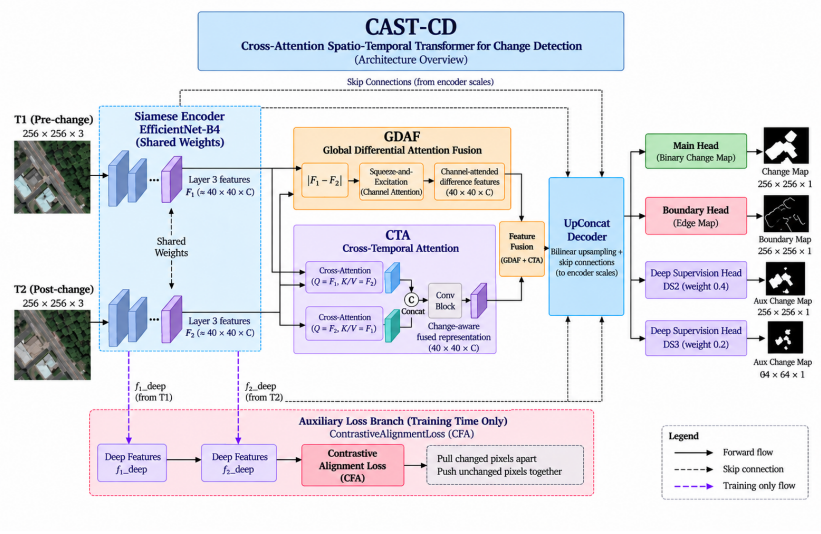
Efficiency vs. Tiny-CD

- **Tiny-CD:** 0.285M params, 3.088 GFLOPs
- **CAST-CD: 0.315M** params, **1.965 GFLOPs**
- $\sim$36% fewer GFLOPs, comparable size

Why CAST-CD?

- Designed specifically for **bi-temporal change detection**.
- Siamese encoder ensures identical feature extraction for T_1 and T_2.
- **Layer 3 only** – rich semantic features, low computation.
- CTA directly models relationships between temporal features instead of relying only on feature subtraction.
- Bidirectional attention captures both:
  - What appeared
  - What disappeared

Architecture



Methodology & Solution Design

CAST-CD: Cross-Attention Siamese Transformer for Change Detection

Methodological Framework:

- **Input:** Co-registered bi-temporal RGB images $T_1, T_2 \in \mathbb{R}^{256\times256\times3}$
- **Feature Extraction:** Siamese EfficientNet-B4, shared weights, Layer 3 features F_1, F_2
- **Core Algorithms:**
  - GDAF – Global Differential Attention Fusion
  - CTA – Bidirectional Cross-Temporal Attention
  - UpConcat decoder with skip connections
- **Multi-task Learning:** change detection + boundary + deep supervision + CFA loss

Datasets & Preprocessing:

- **LEVIR-CD**, **WUHAN-CD**, **DSIFN-CD**
- 256×256 patches, change-biased cropping
- Flips + random $90^\circ/180^\circ/270^\circ$ rotations

Evaluation Metrics:

- Precision, Recall, F1-Score, IoU, Accuracy

Model Efficiency:

- **Params:** 314,531 ($\approx$0.3M) **GFLOPs:** 1.965

Technologies Used in CAST-CD

Core Architectural Concepts Behind the Model

Feature Extraction

- **Siamese Network** – shared-weight twin encoders for T_1, T_2.
- **EfficientNet-B4** – pretrained CNN backbone for semantic features.
- **Layer 3 Only** – rich semantics, low computation.

Feature Interaction

- **GDAF(Global Differential Attention Fusion)** – SE-style channel attention on absolute feature difference.
- **CTA** – bidirectional cross-temporal attention between T_1, T_2.

Decoding & Output

- **UpConcat Decoder** – bilinear upsampling with skip connections.
- **Boundary Head** – sharper change boundaries.
- **Deep Supervision (DS2, DS3)** – auxiliary losses at decoder stages.

Training Strategy

- **Multi-task Learning** – joint change, boundary, contrastive objectives.
- **CFA(Contrastive Feature Alignment** – separates changed vs. unchanged feature embeddings.

GDAF: Differential Feature Attention

$$D = |F_1 - F_2|, \qquad D_{\text{att}} = SE(D)$$

CTA: Bidirectional Cross-Temporal Attention

$$\begin{aligned} A_{12} &= \text{Attention}(Q = F_1, K = F_2, V = F_2) \\ A_{21} &= \text{Attention}(Q = F_2, K = F_1, V = F_1) \\ F_{\text{CTA}} &= \text{Conv}\left([A_{12}; A_{21}]\right) \end{aligned}$$

Experimental Results – WHU-CD

Performance Comparison with Existing Methods

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT-CD	86.64	81.48	98.75	83.98	72.39	10.63	15 ms	3.55 M
U-Net	35.95	93.60	93.74	51.95	34.67	3.13	15 ms	1.35 M
Tiny-CD	94.55	92.79	98.10	93.66	88.08	3.088	736.89 ms	0.285 M
FC-Siam	68.87	85.99	91.98	76.48	61.92	9.453	5724.38 ms	1.35 M
ChangeFormer	91.83	88.02	99.12	89.88	81.63	202.79	81 ms	41.03 M
ChangeStar	88.78	85.31	98.70	87.01	77.00	68.52	19.65 ms	16.77 M
SNUNet	85.60	81.49	98.71	83.50	71.67	54.83	22 ms	12.03 M
STANet	79.37	85.85	98.52	82.32	69.95	16.89	42 ms	16.89 M
MSCDAN	88.82	81.41	98.69	84.95	77.52	9.80	78 ms	11.7 M
CAST-CD	**95.06**	**93.45**	**98.27**	**94.25**	**89.12**	**1.965**	–	**0.315 M**

Analysis in WHU-CD dataset

Experimental Results – LEVIR-CD

Performance Comparison with Existing Methods

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT-CD	89.24	89.37	98.92	89.31	80.68	8.7522	251.6 ms	3.55 M
U-Net	86.91	80.17	98.39	83.40	71.53	3.13	621.3 ms	1.35 M
Tiny-CD	93.14	89.93	99.15	91.51	84.35	3.088	7840.2 ms	0.285 M
FC-Siam	32.98	85.97	90.39	47.68	31.30	9.453	1663.3 ms	1.35 M
ChangeFormer	92.05	88.80	99.04	90.40	82.48	202.79	45.89 ms	41.03 M
ChangeStar	77.33	84.79	98.14	90.82	83.19	29.26	19.65 ms	33.96 M
SNUNet	89.18	87.17	98.82	88.16	78.83	54.83	62 ms	12.03 M
STANet	83.81	88.76	98.66	87.26	77.40	16.89	6 ms	16.92 M
MSCDAN	89.68	84.49	98.81	89.30	86.17	9.80	75 ms	11.7 M
CAST-CD	**87.75**	**88.61**	**98.79**	**88.18**	**78.86**	**1.965**	–	**0.315 M**

Analysis in LEVIR-CD dataset

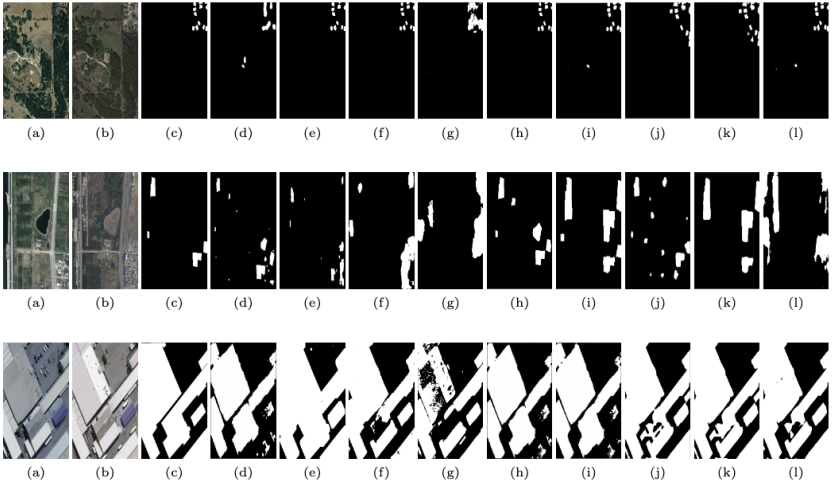
Experimental Results – DSIFN-CD

Performance Comparison with Existing Methods

Model	Precision	Recall	Accuracy	F1	IoU	GFLOPs	Inference	Params
BIT	68.36	70.18	89.41	69.26	52.97	4.35	14270 ms	3.55 M
UNet	72.61	52.73	88.59	61.09	43.98	3.13	4280 ms	1.35 M
Tiny-CD	90.55	90.66	93.79	90.60	82.82	3.088	3130 ms	0.285 M
FC-Siam	64.92	75.54	76.84	69.83	53.65	9.453	82680 ms	1.35 M
ChangeFormer	88.48	84.94	95.56	86.67	76.48	202.79	45.89 ms	41.03 M
ChangeStar	85.90	82.81	88.68	84.33	72.90	29.26	22.38 ms	33.96 M
SNUNet	60.60	72.89	87.34	66.18	49.45	54.83	54 ms	12.03 M
STANet	67.71	61.68	88.49	64.56	47.66	16.89	82 ms	16.93 M
MSCDAN	85.61	84.37	98.69	89.27	84.99	9.80	78.66 ms	11.7 M
CAST-CD	**90.21**	**91.95**	**94.04**	**91.07**	**83.60**	**1.965**	–	**0.315 M**

Analysis in DSIFN-CD dataset

Change Map



(a) T1 **(b)** T2 **(c)** Ground Truth **(d)** BIT **(e)** U-Net **(f)** TinyCD **(g)** FC-Siam **(h)** ChangeFormer **(i)** ChangeStar **(j)** SNUNet **(k)** STANet **(l)** CAST-CD

Preliminary Experimental Verification

Current Results Across Benchmark Datasets

Dataset	Precision	Recall	F1	IoU	Accuracy
WUHAN-CD	0.9506	0.9345	**0.9425**	0.8912	0.9827
DSIFN-CD	0.9021	0.9195	**0.9107**	0.8360	0.9404
LEVIR-CD	0.8775	0.8861	**0.8818**	0.7886	0.9879

Current Implementation

314,531 parameters | **1.965 GFLOPs** | **256×256 input**

Implementation Progress

Core Modules Implemented, Functional & Integrated

Module / Component	Implementation Status	Status
Data Pipeline	Dataset loading, 256×256 patch processing, augmentation and change-biased cropping	Functional
Siamese Encoder	EfficientNet-B4 with shared weights; Layer 3 feature extraction	Functional
GDAF	Absolute feature difference + SE-style channel attention	Functional
CTA	Bidirectional cross-temporal attention ($F_1 \rightarrow F_2,\ F_2 \rightarrow F_1$)	Functional
Decoder	UpConcat decoder with bilinear upsampling and skip connections	Functional
Output Heads	Main change map, Boundary Head, DS2 and DS3	Integrated

Training & Optimization Pipeline

Curriculum Transfer Learning Strategy

Training Configuration

- Batch size: **12**
- Input patches: 256×256
- AMP mixed precision
- Change-biased cropping:
  - 80% centered on changed pixels
- Augmentation:
  - Horizontal / vertical flips
  - Random rotations

Hardware

- Kaggle P100 / T4 GPUs

Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

- **Individual MOOC / Online Certification Selection:**

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	Aditya Suresh	AI Upskilling Certificate: Hands-On Development from Model to App	Qualcomm Academy	15 Hrs
2	Ananthasagar N	IBM Deep Learning Professional Certificate	IBM	19 Hrs
3	Pratyush Yadav	AI and Networks	Udemy	19 Hrs
4	Karthik P	Satellite Imagery, Remote Sensing & Machine Learning Specialization	Coursera	19 Hrs
5	Suriakiran B	Remote Sensing Image Acquisition, Analysis and Applications	Coursera	21 Hrs

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Thank You